
A Shared Standard for Valid Measurement of Generative AI Systems’ Capabilities, Risks, and Impacts

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Abstract

The valid measurement of generative AI (GenAI) systems’ capabilities, risks, and impacts forms the bedrock of our ability to evaluate these systems. We introduce a shared standard for valid measurement that helps place many of the disparate-seeming evaluation practices in use today on a common footing. Our framework, grounded in measurement theory from the social sciences, extends the work of Adcock and Collier [1] in which the authors formalized *valid measurement* of concepts in political science via three processes: *systematizing* background concepts, *operationalizing* systematized concepts via annotation procedures, and *applying* those procedures to instances. We argue that valid measurement of GenAI systems’ capabilities, risks, and impacts, further requires systematizing, operationalizing, and applying not only the entailed *concepts*, but also the *contexts* of interest and the *metrics* used. This involves both *descriptive* reasoning about particular instances and *inferential* reasoning about underlying populations, which is the purview of statistics. By placing many disparate-seeming GenAI evaluation practices on a common footing, our framework enables individual evaluations to be better understood, interrogated for reliability and validity, and meaningfully compared. This is an important step in advancing GenAI evaluation practices toward more formalized and theoretically grounded processes—i.e., toward a science of GenAI evaluations.

1 Introduction

Whether we are interested in what a generative (GenAI) system is capable of, the risks it poses, or the impacts of its deployment, we can gain insight into such inquiries by framing them as *measurement tasks* of the form: *measure the [amount] of a [concept] in [instances] from a [population]*. Examples include evaluating an LLM-based tutoring system’s mathematical reasoning capabilities by measuring its average (amount) performance (concept) on standardized test questions (instances) from the Korean CSAT Math Exam (population); or evaluating the risks posed by a conversational search system by measuring the prevalence (amount) of stereotyping (concept) in the system’s outputs (instances) in the current US deployment (population). This template is also sufficiently expressive to capture many of the impact evaluation tasks described by Solaiman et al. [19]. Recent work has emphasized the need, when carrying out such measurement tasks, to precisely articulate *what* is being measured—i.e., to formalize the *concept* of interest. This includes critical work on dataset diversity [21], benchmark design for stereotyping and other concepts [2, 11], and model memorization [3, 4, 9].

These critiques parallel those made two decades ago by Adcock and Collier [1], who argued that political scientists were devoting insufficient attention to measurement validity—i.e., the question of whether researchers’ reported measurements adequately reflect the concepts they purport to measure. To facilitate this type of reflection, Adcock & Collier introduced a four-level framework that

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formalizes the relationship between a concept of interest and reported measurements for that concept as a structured progression from a *background concept* (the set of meanings and understandings associated with the concept), to a *systematized concept* (precise definitions), to *annotation procedures* (procedures for labeling or scoring instances) that operationalize the systematized concept, to the *application* of those procedures to obtain scores or labels for one or more instances. Validity concerns about concepts then arise as slippage between these four levels, such as a failure of the annotation procedures to capture relevant dimensions of the systematized concept. Using this framework, the above-mentioned critiques of GenAI evaluations can be posed as failures to systematize nebulous background concepts (e.g., stereotyping) before jumping to operationalization via annotation procedures (e.g., instructions that ask crowdworkers to label system outputs as stereotyping or not).

2 Framework Overview

Although we wholeheartedly agree with these critiques, we argue that equal attention should be paid to validity concerns that arise from the under-specification of *amounts*, *populations*, and *instances* entailed in such measurement tasks. In other words, valid measurement of GenAI systems’ capabilities, risks, and impacts requires systematizing, operationalizing, and applying not only *concepts*, but also *amounts*, *populations*, and *instances*. This involves both *descriptive* reasoning about particular instances and *inferential* reasoning about underlying populations, which is the purview of statistics.

To facilitate reflection on such validity concerns, we introduce an extension of Adcock and Collier’s framework [1]. A graphical representation of this framework is depicted in Figure 1. We developed and iteratively refined the framework using numerous examples of GenAI and non-generative AI systems’ capabilities, risks, and impacts, including topic modeling, representational harm measurement for multi-modal models, face verification technologies, automated speech recognition, and others.

Like Adcock & Collier’s framework, our framework includes processes, depicted as backward arrows, for revising and refining earlier levels based on findings, including validity concerns, that arise in later levels. These revision and refinement processes operate not only within each column (e.g., within the “Concept” column, which is adapted from Adcock & Collier’s framework), but also across the columns. We may find, for example, that the variance of our measurements is too high, leading us to revise the sampling design or even the annotation procedures to improve estimator stability.

To better understand the elements of our framework, consider the example of measuring stereotyping in a conversational search system. How we represent an instance (e.g., as a system output vs. as a system output and its corresponding input) may influence whether a system output is labeled as stereotyping. Likewise, if we are interested in the prevalence of stereotyping under typical use post-deployment, but our sampling design (operationalized population) is based on adversarial red teaming, we are likely to vastly overestimate the prevalence. By structuring the choice of metric as first defining a *target parameter* (systematized amount) that is then estimated using statistical estimators (operationalized amount), we can apply established statistical inference methods to optimally trade off between bias and variance, adjust for sample bias in the data set (i.e., correct for mismatches between the systematized population and the sampling design), construct uncertainty intervals, and perform hypothesis tests. In Figure 2, we provide a high-level example of using our framework to measure stereotyping in a hypothetical conversational search engine, ChatSearch.

3 Limitations and Future Work

Although our framework brings structure to the question of *what* should be done when measuring GenAI systems’ capabilities, risks, and impacts, we offer no guidance on *how* to accomplish such measurement tasks. We intend to address this limitation in a future, longer version of this paper.

A major limitation of the framework itself is that it does not help with formulating measurement tasks in the first place, nor does it offer guidance on how to interpret resulting measurements or what decisions those measurements can or should inform. Just as the initial problem formulation can have a big effect on the fairness of an ML model [10, 15–17, 20] or the conclusions drawn about deep-learning optimizer performance [5, 8, 14], what we choose to measure, in what population, and how we quantify it matters a great deal [6, 7, 13]. The revision and refinement processes provide one mechanism for revisiting these decisions, but this falls well short of addressing the full complexity of formulating measurement tasks—an issue that is critically important to evaluations of GenAI systems.

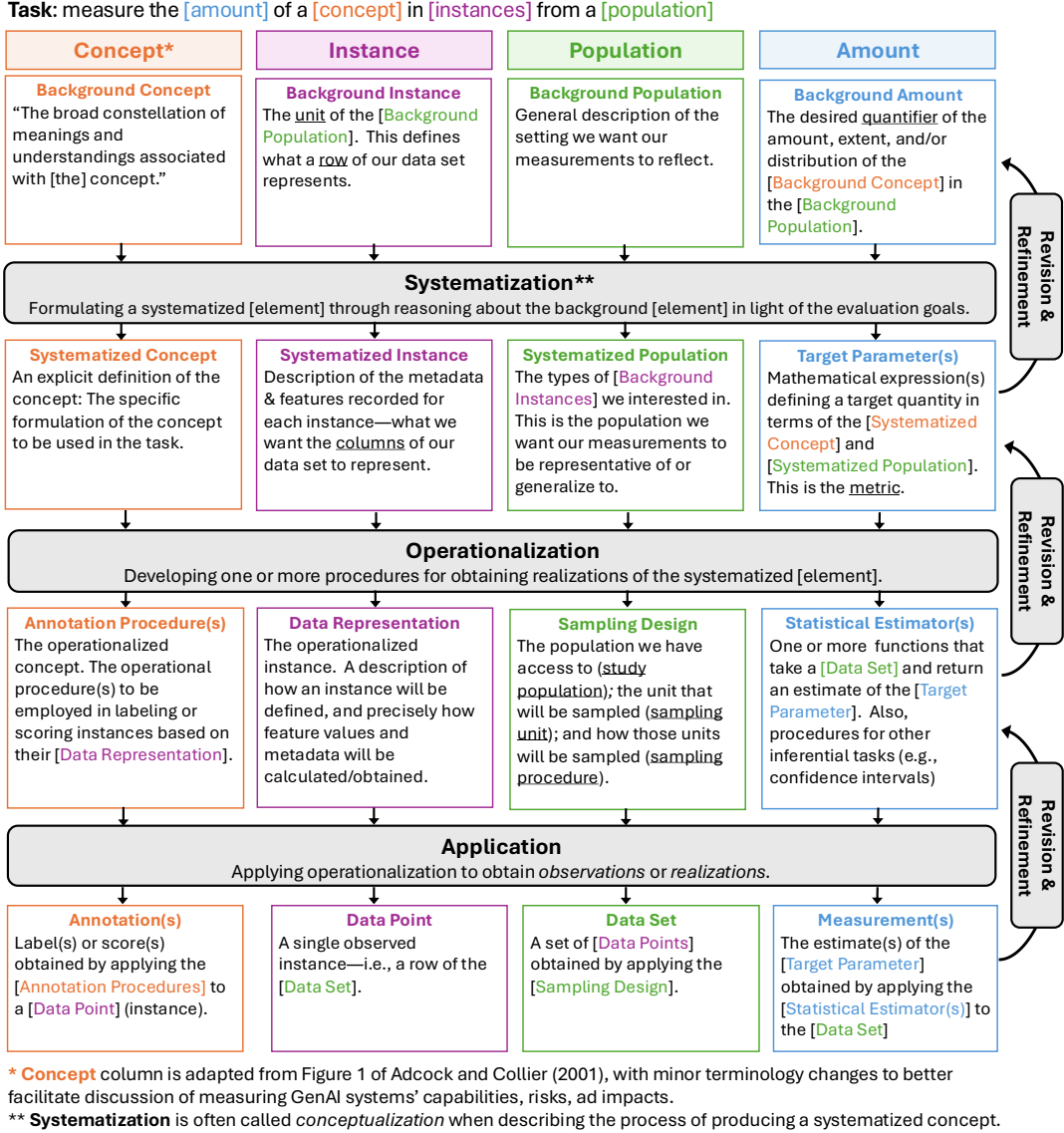


Figure 1: Our proposed framework for measurement tasks of the form: *measure the [amount] of a [concept] in [instances] from a [population]*. The figure shows how the four elements that make up such tasks—amounts, concepts, instances, and populations—are formalized through the sequential processes of *systematization*, *operationalization* and *application*. Elements in earlier levels (rows) can be revised and refined based on findings, including validity concerns, that arise in later levels.

4 Conclusion

Our framework helps place many of the disparate-seeming GenAI evaluation practices in use today on a common footing. This enables individual evaluations to be better understood, interrogated for reliability and validity, and meaningfully compared. In this way, the framework is intended to serve as a shared standard for valid measurement of GenAI systems’ capabilities, risks, and impacts. For example, using it allows us to (1) more easily identify and remedy validity concerns; and (2) compare different measurement tasks by identifying precisely where within the framework they diverge. In other words, our framework enables us not only to “evaluate evaluations” but also to improve how evaluations of GenAI systems are designed in the first place, thereby helping advance such evaluations from their current state of disparate-seeming and ad hoc practices [12, 18] toward more formalized and theoretically grounded processes—i.e., toward a science of GenAI evaluations.

Evaluation goals: The product team for ChatSearch, an LLM-enabled conversational search engine, wants to assess their system for representational harms. Red teaming results show that the system can output responses that stereotype or demean, and they want to quantify the extent of the problem.

Task: measure the [prevalence] of [stereotyping] for [ChatSearch responses] to user queries from [typical user interactions]

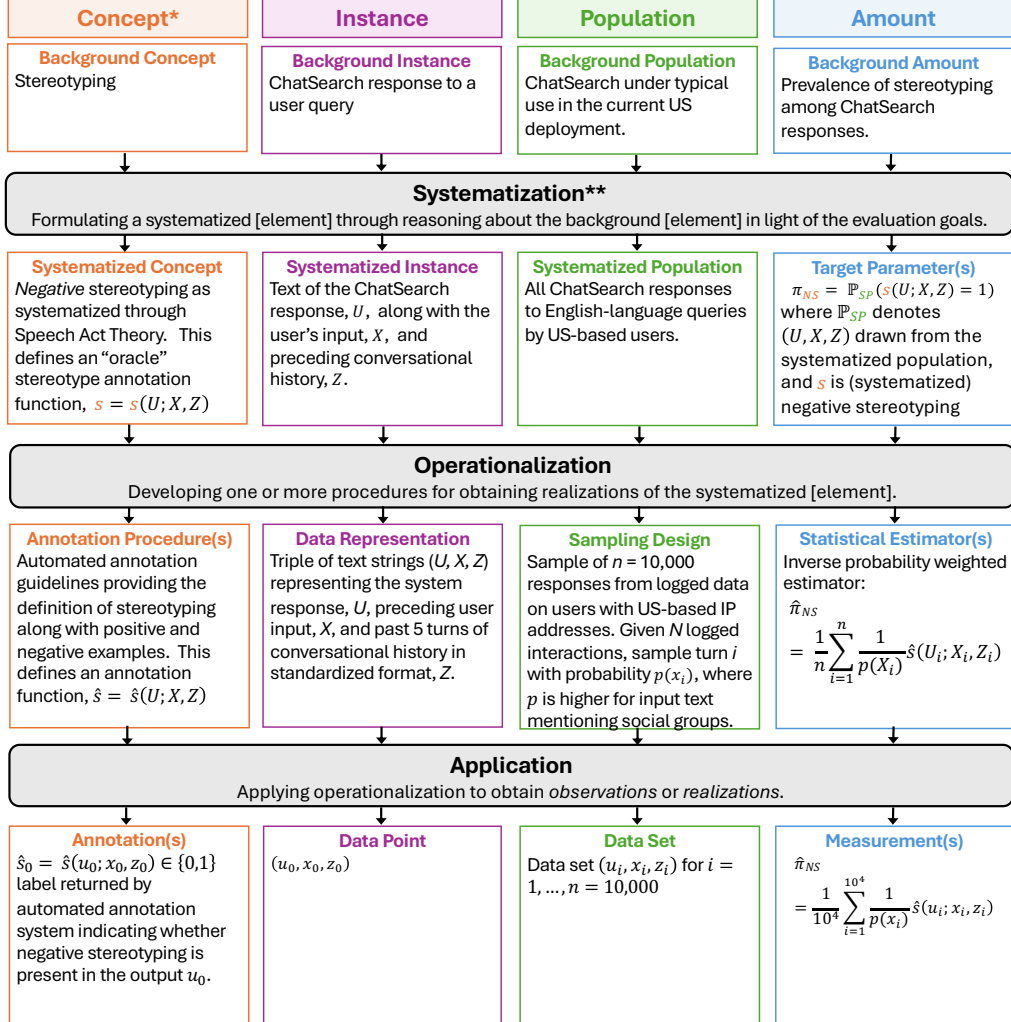


Figure 2: A high-level example of using our framework in a hypothetical evaluation of a conversational search engine, ChatSearch. Each cell provides an overview of what a complete measurement procedure instantiated using the framework could look like. Note that a full instantiation would require providing considerable additional information. For example, a fully systematized complex concept or the full description of a complex sampling design might require several pages of exposition.

References

- [1] Robert Adcock and David Collier. Measurement validity: A shared standard for qualitative and quantitative research. *American political science review*, 95(3):529–546, 2001.
- [2] Su Lin Blodgett, Gilsinia Lopez, Alexandra Olteanu, Robert Sim, and Hanna Wallach. Stereotyping norwegian salmon: An inventory of pitfalls in fairness benchmark datasets. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 1004–1015, 2021.
- [3] Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. Quantifying memorization across neural language models. *arXiv preprint arXiv:2202.07646*, 2022.

- [4] A Feder Cooper and James Grimmermann. The Files are in the Computer: Copyright, Memorization, and Generative AI. *arXiv preprint arXiv:2404.12590*, 2024.
- [5] A. Feder Cooper, Yucheng Lu, Jessica Forde, and Christopher M De Sa. Hyperparameter Optimization Is Deceiving Us, and How to Stop It. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 3081–3095. Curran Associates, Inc., 2021.
- [6] A. Feder Cooper, Katherine Lee, Madiha Zahrah Choksi, Solon Barocas, Christopher De Sa, James Grimmermann, Jon Kleinberg, Siddhartha Sen, and Baobao Zhang. Arbitrariness and Social Prediction: The Confounding Role of Variance in Fair Classification. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(20):22004–22012, March 2024.
- [7] Amanda Coston, Alan Mishler, Edward H Kennedy, and Alexandra Chouldechova. Counterfactual risk assessments, evaluation, and fairness. In *Proceedings of the 2020 conference on fairness, accountability, and transparency*, pages 582–593, 2020.
- [8] Jesse Dodge, Suchin Gururangan, Dallas Card, Roy Schwartz, and Noah A. Smith. Show Your Work: Improved Reporting of Experimental Results. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2185–2194, Hong Kong, China, November 2019. Association for Computational Linguistics.
- [9] Daphne Ippolito, Florian Tramèr, Milad Nasr, Chiyuan Zhang, Matthew Jagielski, Katherine Lee, Christopher A. Choquette-Choo, and Nicholas Carlini. Preventing Verbatim Memorization in Language Models Gives a False Sense of Privacy, 2023. URL <https://arxiv.org/abs/2210.17546>.
- [10] Benjamin Laufer, Thomas Gilbert, and Helen Nissenbaum. Optimization’s neglected normative commitments. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, pages 50–63, 2023.
- [11] Yu Lu Liu, Su Lin Blodgett, Jackie Cheung, Q. Vera Liao, Alexandra Olteanu, and Ziang Xiao. ECBD: Evidence-centered benchmark design for NLP. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16349–16365, Bangkok, Thailand, August 2024. Association for Computational Linguistics. URL <https://aclanthology.org/2024.acl-long.861>.
- [12] Nestor Maslej, Loredana Fattorini, Raymond Perrault, Vanessa Parli, Anka Reuel, Erik Brynjolfsson, John Etchemendy, Katrina Ligett, Terah Lyons, James Manyika, Juan Carlos Niebles, Yoav Shoham, Russell Wald, and Jack Clark. The AI index 2024 annual report. AI Index Steering Committee, Institute for Human-Centered AI, Stanford University, April 2024.
- [13] Dylan Mulvin. *Proxies: The cultural work of standing in*. MIT Press, 2021.
- [14] Johan Obando-Ceron, João G. M. Araújo, Aaron Courville, and Pablo Samuel Castro. On the consistency of hyper-parameter selection in value-based deep reinforcement learning, 2024. URL <https://arxiv.org/abs/2406.17523>.
- [15] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464):447–453, 2019.
- [16] Samir Passi and Solon Barocas. Problem formulation and fairness. In *Proceedings of the conference on fairness, accountability, and transparency*, pages 39–48, 2019.
- [17] Shangshu Qian, Hung Viet Pham, Thibaud Lutellier, Zeou Hu, Jungwon Kim, Lin Tan, Yaoliang Yu, Jiahao Chen, and Sameena Shah. Are my deep learning systems fair? an empirical study of fixed-seed training. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, 2021. URL <https://openreview.net/forum?id=kLWGdQYsmC5>.
- [18] Kevin Roose. A.I. has a measurement problem. *The New York Times*, April 2024. URL <https://www.nytimes.com/2024/04/15/technology/ai-models-measurement.html>. Accessed: 2024-09-05.
- [19] Irene Solaiman, Zeerak Talat, William Agnew, Lama Ahmad, Dylan Baker, Su Lin Blodgett, Canyu Chen, Hal Daumé III, Jesse Dodge, Isabella Duan, et al. Evaluating the social impact of generative ai systems in systems and society. *arXiv preprint arXiv:2306.05949*, 2023.
- [20] Jamelle Watson-Daniels, Solon Barocas, Jake M Hofman, and Alexandra Chouldechova. Multi-target multiplicity: Flexibility and fairness in target specification under resource constraints. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, pages 297–311, 2023.
- [21] Dora Zhao, Jerone TA Andrews, Orestis Papakyriakopoulos, and Alice Xiang. Position: Measure dataset diversity, don’t just claim it. *arXiv preprint arXiv:2407.08188*, 2024.